RUDRAKSH SYAL

Technical Lead | Agentic AI & LLM Platforms

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PROFESSIONAL PROFILE

Technical lead with 11+ years in Python, ML, and data science, leading a team of two data scientists since May 2024 and owning architecture and code reviews. Built and shipped an LLM tool-calling layer on Claude via AWS Bedrock, exposed as an API/service consumed by other teams and powering a live V1 dashboard used daily by revenue leadership and fleet pricing managers across 7 U.S. regions (~240 requests/day, 85–90% recommendation acceptance). Also built hybrid RAG over 4,000+ contracts with evaluation and human-in-the-loop review.

TECHNICAL SKILLS

Agentic AI & LLMs: Tool/function calling, multi-step agent orchestration, RAG, hybrid/vector search, embeddings, MCP, LangChain/LangGraph, OpenAI Agents SDK, Claude via AWS Bedrock, OpenAI, Gemini, Llama 3.1 8B

Evaluation & Observability: LLM-as-a-judge, golden sets, Precision@K/Recall@K, guardrails, LangSmith, OpenTelemetry, human-in-the-loop review

Cloud, Data & Platform: AWS (Bedrock, SageMaker, OpenSearch Serverless), Docker, Kubernetes, PostgreSQL, Snowflake, Databricks, Azure, MLflow, Airflow, Python, SQL, Spark

ML & Statistics: XGBoost, Random Forest, forecasting, optimization, causal inference (BSTS, synthetic control), A/B testing, NLP

PROFESSIONAL EXPERIENCE

Data Science Lead | Hana E Solutions **Oct 2026–Present**

Remote

- **Agentic AI workflows:** Leading agentic AI workflow design from scratch for two clients, mainly in pricing; both at proof-of-concept stage.

Data Science Lead | Vista **May 2024–Jul 2026**

San Jose, CA (Remote)

- **Team leadership:** Led two data scientists, set technical direction, reviewed code and model designs, and owned architecture for pricing and agentic AI systems.
- **Agentic platform architecture:** Led design of a custom Python LLM tool-calling layer (no framework) on Claude Sonnet, Haiku, and Opus via AWS Bedrock, on top of the Dynamic Pricing Engine. The LLM calls forecasting and Snowflake tools, explains pricing drivers, and a pricing manager approves or rejects each recommendation.
- **API/service & live dashboard:** Exposed the layer as an API/service consumed by other teams, feeding a live V1 dashboard that revenue leadership and fleet pricing managers monitor daily.
- **Parallel run to adoption:** Ran a six-quarter program across 7 U.S. regions. A Q6 parallel run won buy-in, and the layer went live in Q7 at ~240 requests a day across 3 fleets. Rejections fell from ~30% to 10–15%.
- **Evaluation & observability:** Tied LLM-as-a-judge evaluations to approve/reject outcomes, with acceptance rate as the success metric. Wrote guardrails in custom Python and traced runs with LangSmith.
- **Hybrid RAG over contracts:** Built hybrid retrieval over 4,000+ contracts with Bedrock Knowledge Bases, OpenSearch Serverless vector search, and a SageMaker-hosted Llama 3.1 8B, evaluated on golden question–clause sets (Precision@K/Recall@K).

- **Dynamic Pricing Engine:** Led two data scientists to replace static pricing bands with a supply-demand equilibrium engine across 7 U.S. regions, using six supply submodels and a bounded Brentq yield solver. One 43-aircraft fleet saw 28% fewer off-fleet bookings in month one.

Senior Data Scientist | Vista **Oct 2021–Apr 2024**

San Jose, CA (Remote)

- **LLM extraction to production:** Structured refund terms for ~600 operators into cancellation-policy badges with product and engineering. 84.7% were badge-ready automatically; uncertain cases went to manual review.
- **EU cost model:** Built a gradient-boosted cost model, validated in shadow mode before relaunch. Margin went from −8% to +17%, and loss-making bookings fell from 55% to 20%.
- **Forecasting & propensity:** Raised booking forecast accuracy for 14 fleet categories from 71% to 78%. Built XGBoost lead-propensity scores integrated into Salesforce.

Senior Data Scientist | 84.51° (Kroger) **Jul 2019–Oct 2021**

Cincinnati, OH

- **Causal measurement:** Built BSTS and synthetic-control measurement for Quaker endcaps (4% lift vs. a 10% benchmark), which informed the decision to pause an Eastern-division rollout.
- **Space optimization:** Optimized bakery footage with Random Forest and integer programming. Nine pilot stores recorded 14% higher early sales (directional).

Data Scientist | Practo Technologies **Jun 2015–Apr 2018**

Bangalore, India

- **Analytics apps & NLP:** Built R/Shiny and Flask reporting apps that replaced a $4K/month-per-person vendor. Used sentiment NLP and A/B tests to find conversion drop-offs.

PROJECTS

- **Dynamic Pricing Engine** (personal rebuild on anonymised data)
  github.com/rudrakshsyal/dynamic_pricing_engine
  An OpenAI Agents SDK layer writes tool-bound revenue briefs, checked for citations and numbers. A budget-capped GPT-5.5 research agent was benchmarked against production-style ML forecasts; results were mixed, so the ML forecasts were kept.
- **Sidekick Agent** (personal project)
  github.com/rudrakshsyal/the_sidekick_agent
  A LangChain/LangGraph agent with MCP tools (Playwright browser, sandboxed filesystem), an evaluator loop with up to 3 retries, PII and model-call-limit guardrails, and human-in-the-loop approval.

LEADERSHIP, PATENT & EDUCATION

Leadership: Led two data scientists (May 2024–Jul 2026); started cross-functional learning sessions for 60+ colleagues.

Patent: U.S. Patent No. 12,314,757 B1 (2025): Efficient resource scheduling using adaptive scheduling criteria (co-inventor).

Purdue University — M.S., Business Analytics & Data Science (2019)

BITS Pilani — B.E. (Hons.) and M.Sc., Applied Mathematics (2015)